

Capstone Project and Test Summary

Demo Web Shop Automation Testing using Playwright

Field	Details
Submitted By	Akshit
Project Name	Playwright Capstone Project
Application Under Test	Demo Web Shop
Automation Framework	Playwright with JavaScript
GitHub Repository	https://github.com/Akshit1984/Playwright_Capstone-Project
Report Date	05 June 2026

Overview

The Application Under Test (AUT) is Demo Web Shop, a public e-commerce web application that simulates typical retail workflows such as product browsing, authentication, cart access, checkout information, customer support, wishlist, comparison, and account-related navigation.

This report documents the completed automation testing capstone project. It covers the testing objectives, requirement analysis, functional and non-functional requirements, automation strategy, test design methodology, module coverage, execution approach, reporting setup, and final project outcome.

Repository: https://github.com/Akshit1984/Playwright_Capstone-Project

Goals

Business Objectives

- Validate critical e-commerce user journeys before release or demonstration.
- Ensure users can access core pages such as login, product categories, cart, checkout, account, and support information.
- Confirm that add-on features such as wishlist, comparison, and customer support pages are covered separately and clearly.

Quality Engineering Objectives

- Create maintainable Playwright automation for repetitive regression checks.
- Provide cross-browser verification through Chromium, Firefox, and WebKit.
- Improve execution speed using parallel workers and reusable fixture design.
- Generate readable reports for trainer review using Playwright and Allure reporting.

Specifications

In-Scope Functional Modules

Testing covers six core modules and two add-on modules as per the updated project instruction.

Type	Service Module	Test Cases	Test ID Prefix	Primary Coverage
Core	Authentication	15	AUTH	Login, registration, password recovery, protected page redirect, header auth links
Core	Product and Search	15	PROD	Home page, category navigation, products, details, search results
Core	Cart	15	CART	Empty cart state, cart links, add-to-cart controls, order progress labels
Core	User Profile and Account	15	USER	Account pages, order history, addresses, registration fields, account links
Core	Address and Shipping	15	ADDR	Shipping returns, delivery text, free shipping text, availability, sitemap
Core	Checkout and Payment	15	PAY	Checkout access, payment labels, gift card fields, product price information
Add on	Add on - Wishlist and Compare	15	WISH	Wishlist, compare products, recently viewed, new products, email-a-friend
Add on	Add on - Customer Support and Information	15	INFO	Contact us, sitemap, privacy, conditions, about, news, blog, newsletter, social links

Out of Scope

- Backend database validation is not performed because Demo Web Shop is a public hosted demo application.
- Payment gateway transaction processing is not validated with real payment data.
- Load testing and security penetration testing are not included in this automation scope.

Milestone 1 - Analysis and Planning

1.1 Introduction

The planning phase defines what should be tested, why it should be tested, and how the testing work should be structured. For this project, the application was analyzed as an e-commerce system with customer-facing modules and supporting informational pages.

1.2 Requirement Analysis

The requirements were grouped into Functional Requirements (FR) and Non-Functional Requirements (NFR). Functional requirements describe what the application should do. Non-functional requirements describe how well the application and testing solution should behave across quality attributes such as compatibility, reliability, maintainability, and reporting.

1.2.1 Functional Requirements (FR)

FR ID	Module	Requirement Description	Priority	Mapped Tests
FR-01	Authentication	The application should provide visible login, registration, and password recovery pages.	High	AUTH-001 to AUTH-015
FR-02	Product and Search	The application should allow users to browse categories, open product details, and search for products.	High	PROD-001 to PROD-015
FR-03	Cart	The application should expose shopping cart access, empty-cart behavior, product cart controls, and checkout journey labels.	High	CART-001 to CART-015
FR-04	User Profile and Account	Guest users should be redirected to login for account, orders, and address pages, while account links remain discoverable.	High	USER-001 to USER-015
FR-05	Address and Shipping	Shipping and address-related information should be visible through product pages, sitemap links, and shipping pages.	Medium	ADDR-001 to ADDR-015
FR-06	Checkout and Payment	Checkout should block empty carts and display payment, confirmation, completion, gift-card, and price information correctly.	High	PAY-001 to PAY-015
FR-07	Add on - Wishlist and Compare	Users should be able to access wishlist, compare products, recently viewed products, new products, and email-a-friend pages.	Medium	WISH-001 to WISH-015
FR-08	Add on - Customer Support and Information	Users should be able to access contact, sitemap, privacy, conditions, about, news, blog, newsletter, and social information.	Medium	INFO-001 to INFO-015

Milestone 1 - Non-Functional Requirements

1.2.2 Non-Functional Requirements (NFR)

NFR ID	Quality Attribute	Requirement Description	Priority	Implementation Evidence
NFR-01	Browser Compatibility	The automated checks should run on Chromium, Firefox, and WebKit to validate major browser-engine behavior.	High	Configured through Playwright projects.
NFR-02	Performance of Test Suite	The suite should use parallel execution and reusable fixtures to reduce repeated browser setup overhead.	High	Workers and worker-scoped fixture optimization applied.
NFR-03	Maintainability	Tests should be modular, readable, and easy to update when page behavior changes.	High	Page Object Model and module-wise folders implemented.
NFR-04	Reliability	Assertions should validate stable visible behavior and avoid unnecessary waits or fragile dependency on generated data.	High	Visibility, title, URL, and text assertions used.
NFR-05	Portability	The project should run on another PC after installing Node dependencies and Playwright browsers.	High	package.json scripts and setup guidance included.
NFR-06	Reportability	Execution results should be reviewable through Playwright HTML and Allure reports.	Medium	Report scripts and GitHub Actions workflow configured.
NFR-07	Traceability	Each test should map back to a service module and a readable test ID.	Medium	AUTH, PROD, CART, USER, ADDR, PAY, WISH, INFO prefixes used.
NFR-08	Scalability	Additional modules should be addable without disturbing existing tests.	Medium	Separate folders and page objects support extension.

1.3 Test Strategy

A hybrid quality strategy was followed. The project uses automated regression tests for repeated browser validation and manual review for report quality, project structure, and trainer-facing deliverables. The automation suite is the primary evidence for functional coverage.

- Smoke validation: quick browser-level checks using selected commands.
- Regression validation: full test execution across the configured browser projects.
- Documentation validation: final report and testing plan maintained in the docs folder.
- Report validation: Playwright HTML and Allure reporting configured for execution review.

Milestone 1 - RTM Structure

1.4 Requirements Traceability Matrix

The Requirements Traceability Matrix (RTM) links project requirements to test coverage. This ensures that each major functional requirement is connected to a test module and a test ID range.

Requirement ID	Service Module	Test Folder	Test Range	Coverage Status
FR-01	Authentication	tests/authentication	AUTH-001 to AUTH-015	Covered
FR-02	Product and Search	tests/product	PROD-001 to PROD-015	Covered
FR-03	Cart	tests/cart	CART-001 to CART-015	Covered
FR-04	User Profile and Account	tests/user-profile	USER-001 to USER-015	Covered
FR-05	Address and Shipping	tests/address-shipping	ADDR-001 to ADDR-015	Covered
FR-06	Checkout and Payment	tests/checkout-payment	PAY-001 to PAY-015	Covered
FR-07	Add on - Wishlist and Compare	tests/add-on-wishlist-and-compare	WISH-001 to WISH-015	Covered
FR-08	Add on - Customer Support and Information	tests/add-on-customer-support-and-information	INFO-001 to INFO-015	Covered

Milestone 2 - Static and Manual Testing

2.1 Static Review

Static review was performed on the project structure, naming conventions, test grouping, report outputs, and script configuration. The add-on modules were renamed so that their folder names, spec file names, and Playwright test titles show the add-on classification clearly.

Review Area	Observation	Action Taken
Folder structure	Tests were grouped by module.	Add-on folders renamed with add-on prefix.
Test naming	Add-on tests needed clearer names.	Test titles updated to include add-on service name.
Docs folder	Extra generated files were present.	Cleaned to final report and testing plan only.
Execution speed	Earlier runs were slow.	Parallel execution and reusable fixtures configured.

2.2 Manual Verification Scope

- Open key Demo Web Shop pages and verify that expected UI sections are available.
- Review generated reports and confirm the test names are understandable.
- Check that GitHub repository link is present in final documentation.
- Confirm the docs folder only contains the final report and testing plan.

Milestone 3 - Playwright Automation Testing

3.1 Automation Framework

Playwright was selected because it supports modern browser automation, fast execution, auto-waiting, strong assertions, and cross-browser testing from the same test suite. The project uses JavaScript and the Playwright Test runner.

3.2 Page Object Model Theory

The Page Object Model is a design pattern that separates page interaction logic from test case logic. It improves maintainability by placing reusable navigation and verification methods inside page object classes. If a page changes, updates can be made in the page object instead of every test file.

3.3 Fixture Design Theory

Fixtures provide reusable setup for tests. In this project, fixtures create and provide page object instances to test cases. This reduces duplication and keeps spec files focused on test intent rather than setup details.

Milestone 3 - Test Design Methodology

The automated test cases were designed using assertion-driven validation. Each test checks one expected behavior, such as a page opening, a field being visible, a heading being present, or a link being available. This makes failures easier to understand and trace back to a specific requirement.

- Requirement-based design: tests map to functional modules and FR IDs.
- Positive page availability checks: important routes are opened and verified.
- UI element checks: inputs, buttons, links, and labels are validated.
- Navigation checks: protected pages, footer links, sitemap links, and support pages are verified.
- Cross-browser design: the same scenarios are executed across Chromium, Firefox, and WebKit.

Design Principle	How It Was Applied
Modularity	Each service area has its own folder and page object.
Readability	Test IDs and scenario names describe the expected behavior.
Traceability	Prefixes such as AUTH, PROD, CART, USER, ADDR, PAY, WISH, and INFO connect tests to modules.
Maintainability	Common behavior is reused through fixtures and page objects.

Milestone 4 - Test Case Summary

4.1 Overall Test Case Distribution

Module Type	Module	Folder	Test Cases	Total Browser Executions
Core	Authentication	tests/authentication	15	45
Core	Product and Search	tests/product	15	45
Core	Cart	tests/cart	15	45
Core	User Profile and Account	tests/user-profile	15	45
Core	Address and Shipping	tests/address-shipping	15	45
Core	Checkout and Payment	tests/checkout-payment	15	45
Add on	Add on - Wishlist and Compare	tests/add-on-wishlist-and-compare	15	45
Add on	Add on - Customer Support and Information	tests/add-on-customer-support-and-information	15	45

4.2 Execution Count

There are 120 test cases. Since the suite is configured for three browser projects, Playwright discovers 360 browser-specific tests.

Metric	Value
Total service modules	8
Core modules	6
Add-on modules	2
Total test cases	120
Browser projects	3
Discovered tests	360

Test Cases - Authentication

Core

Folder: tests/authentication

15 test cases

Scenario Groups

- Login page availability
- Login form controls
- Password recovery path
- Registration page fields
- Protected page guest redirects

Detailed Test Cases

Test ID	Scenario
AUTH-001	Login page opens with correct title and heading
AUTH-002	Login page shows email and password fields
AUTH-003	Login page shows login button
AUTH-004	Login page shows remember me and forgot password options
AUTH-005	Login page shows registration option for new customer
AUTH-006	Password recovery page opens correctly
AUTH-007	Password recovery page has email field and recover button
AUTH-008	Register page opens correctly
AUTH-009	Register page shows gender choices
AUTH-010	Register page shows personal detail fields
AUTH-011	Register page shows password fields
AUTH-012	Register button is available
AUTH-013	Protected account page redirects guest user to login
AUTH-014	Protected orders page redirects guest user to login
AUTH-015	Header exposes login and register links

Test Cases - Product and Search

Core

Folder: tests/product

15 test cases

Scenario Groups

- Home page content
- Top navigation categories
- Product grid pages
- Product detail pages
- Search results

Detailed Test Cases

Test ID	Scenario
PROD-001	Home page shows welcome content and featured products
PROD-002	Top navigation exposes catalog categories
PROD-003	Books category shows product cards
PROD-004	Computers category shows subcategories
PROD-005	Electronics category shows subcategories
PROD-006	Apparel and shoes category shows products
PROD-007	Digital downloads category opens
PROD-008	Jewelry category opens with product cards
PROD-009	Gift cards category opens with products
PROD-010	Build your own computer product details are visible
PROD-011	Laptop product details are visible
PROD-012	Smartphone product details are visible
PROD-013	Book product page shows price and shipping text
PROD-014	Category page has sort and page size controls
PROD-015	Search page returns computer products

Test Cases - Cart

Core

Folder: tests/cart

15 test cases

Scenario Groups

- Empty cart state
- Cart link visibility
- Product cart controls
- Cart footer links
- Checkout journey labels

Detailed Test Cases

Test ID	Scenario
CART-001	Empty cart page opens
CART-002	Empty cart message is shown
CART-003	Header shopping cart link is visible
CART-004	Cart footer link is visible
CART-005	Book product has add to cart controls
CART-006	Laptop product has add to cart controls
CART-007	Computer product has add to cart controls
CART-008	Gift card product has add to cart controls
CART-009	Apparel product has cart controls
CART-010	Jewelry product has cart controls
CART-011	Cart page still shows search controls
CART-012	Cart page shows order progress steps
CART-013	Cart page shows information footer links
CART-014	Cart page keeps footer follow links available
CART-015	Cart page has checkout journey labels

Test Cases - User Profile and Account

Core

Folder: tests/user-profile

15 test cases

Scenario Groups

- Guest redirects
- Account footer links
- Register form fields
- Password recovery support
- Sitemap account links

Detailed Test Cases

Test ID	Scenario
USER-001	My account page requires login for guest
USER-002	Order history page requires login for guest
USER-003	Addresses page requires login for guest
USER-004	Login page supports returning customer section
USER-005	Footer exposes my account link
USER-006	Footer exposes orders link
USER-007	Footer exposes addresses link
USER-008	Register page shows account creation form
USER-009	Register page shows personal details fields
USER-010	Register page shows email field
USER-011	Register page shows password setup fields
USER-012	Login page provides register button for new user
USER-013	Login page provides forgot password link
USER-014	Password recovery page supports account recovery
USER-015	Sitemap includes account links

Test Cases - Address and Shipping

Core

Folder: tests/address-shipping

15 test cases

Scenario Groups

- Shipping returns page
- Free shipping labels
- Delivery date text
- Stock availability
- Shipping sitemap links

Detailed Test Cases

Test ID	Scenario
ADDR-001	Shipping returns page opens
ADDR-002	Shipping returns page has customer service context
ADDR-003	Guest addresses page redirects to login
ADDR-004	Book product displays free shipping
ADDR-005	Science product displays free shipping
ADDR-006	Health book product displays delivery date
ADDR-007	Laptop product shows in-stock availability
ADDR-008	Smartphone product shows in-stock availability
ADDR-009	Blue jeans product shows in-stock availability
ADDR-010	Shopping cart shows address step
ADDR-011	Shopping cart shows shipping step
ADDR-012	Footer has shipping returns link
ADDR-013	Sitemap includes shipping returns link
ADDR-014	Cart page has checkout progress section
ADDR-015	Shipping page keeps information links available

Test Cases - Checkout and Payment

Core

Folder: tests/checkout-payment

15 test cases

Scenario Groups

- Empty cart checkout block
- Payment step labels
- Gift card fields
- Fixed price checks
- Product price information

Detailed Test Cases

Test ID	Scenario
PAY-001	Empty cart keeps user on shopping cart page
PAY-002	Empty cart blocks checkout with empty cart message
PAY-003	Cart page is available before checkout
PAY-004	Cart page shows payment step
PAY-005	Cart page shows confirm step
PAY-006	Cart page shows complete step
PAY-007	Gift card product has recipient name field
PAY-008	Gift card product has recipient email field
PAY-009	Gift card product has sender fields
PAY-010	Gift card product has message field
PAY-011	Gift card product has add to cart button
PAY-012	Gift card product shows fixed price
PAY-013	Book product shows price information
PAY-014	Laptop product shows price information
PAY-015	Smartphone product shows price information

Test Cases - Add on - Wishlist and Compare

Add on

Folder: tests/add-on-wishlist-and-compare

15 test cases

Scenario Groups

- Wishlist page
- Wishlist empty message
- Compare products page
- Recently viewed/new products
- Email-a-friend form

Detailed Test Cases

Test ID	Scenario
WISH-001	Wishlist page opens
WISH-002	Empty wishlist message is shown
WISH-003	Header wishlist link is visible
WISH-004	Gift card product has wishlist button
WISH-005	Computer product has compare button
WISH-006	Laptop product has compare button
WISH-007	Compare products page opens
WISH-008	Recently viewed products page opens
WISH-009	New products page opens
WISH-010	Product page exposes email-a-friend action
WISH-011	Email-a-friend page opens
WISH-012	Email-a-friend form has friend email field
WISH-013	Email-a-friend form has message and send button
WISH-014	Compare products footer link is available
WISH-015	Wishlist footer link is available

Test Cases - Add on - Customer Support and Information

Add on

Folder: tests/add-on-customer-support-and-information

15 test cases

Scenario Groups

- Contact us form
- Sitemap page
- Privacy and conditions pages
- About/news/blog pages
- Newsletter and social links

Detailed Test Cases

Test ID	Scenario
INFO-001	Contact us page opens
INFO-002	Contact us page has full name field
INFO-003	Contact us page has email field
INFO-004	Contact us page has enquiry textarea
INFO-005	Contact us page has submit button
INFO-006	Sitemap page opens
INFO-007	Privacy notice page opens
INFO-008	Conditions of use page opens
INFO-009	About us page opens
INFO-010	News page opens
INFO-011	Blog page opens
INFO-012	Footer information links are visible
INFO-013	Footer customer service links are visible
INFO-014	Newsletter subscription controls are visible
INFO-015	Footer social links are visible

Milestone 5 - Browser and Execution Plan

5.1 Browser Coverage

Browser Project	Purpose
Chromium	Validates Chrome/Edge-like browser behavior.
Firefox	Validates Mozilla browser engine behavior.
WebKit	Validates Safari-like browser engine behavior.

5.2 Execution Commands

The project can be executed from the terminal using npm scripts. The final report avoids source-code snippets, but includes command-level execution information because it is part of the testing process.

Command	Purpose
npm install	Install project dependencies.
npx playwright install	Install Playwright browser binaries.
npm run test:list	List discovered tests without execution.
npm test	Run the configured Playwright suite.
npm run test:chromium	Run tests on Chromium only.
npm run test:allure	Run tests and generate Allure result files.

Milestone 6 - Reporting and CI/CD

6.1 Playwright Report

Playwright HTML report is useful for local debugging because it shows test status, duration, errors, and browser project details.

6.2 Allure Report

Allure report provides a cleaner dashboard-style view for trainers and reviewers. It supports suites, categories, history, duration, and a presentable test execution summary.

6.3 GitHub Pages Hosting

The project includes a GitHub Actions workflow for publishing the Allure report to GitHub Pages. This helps make test results accessible from the repository after execution.

Project Repository: https://github.com/Akshit1984/Playwright_Capstone-Project

Risk and Mitigation Plan

Risk ID	Risk Description	Level	Mitigation
R-01	External demo site slowness may increase execution time.	Medium	Use reasonable timeouts and parallel workers.
R-02	Public demo site content may change.	Medium	Keep checks focused on stable headings, links, and labels.
R-03	Cross-browser behavior may vary.	Medium	Run tests on Chromium, Firefox, and WebKit.
R-04	Report generation can create extra files.	Low	Docs folder cleaned to final deliverables only.

Defect Management Approach

When a test fails, the defect should be reviewed using the failed test title, module name, browser project, screenshot or report evidence, and expected versus actual result. This helps determine whether the issue is an application defect, a test script issue, a browser-specific behavior, or a temporary environment problem.

Field	Purpose
Defect ID	Unique identifier for tracking.
Module	Affected service module.
Browser	Browser where the issue appeared.
Steps to Reproduce	Short reproduction flow from the test scenario.
Expected Result	Behavior defined by the requirement.
Actual Result	Observed failure behavior.
Severity	Business impact of the issue.
Status	Open, in progress, fixed, retest, or closed.

Final Result Summary

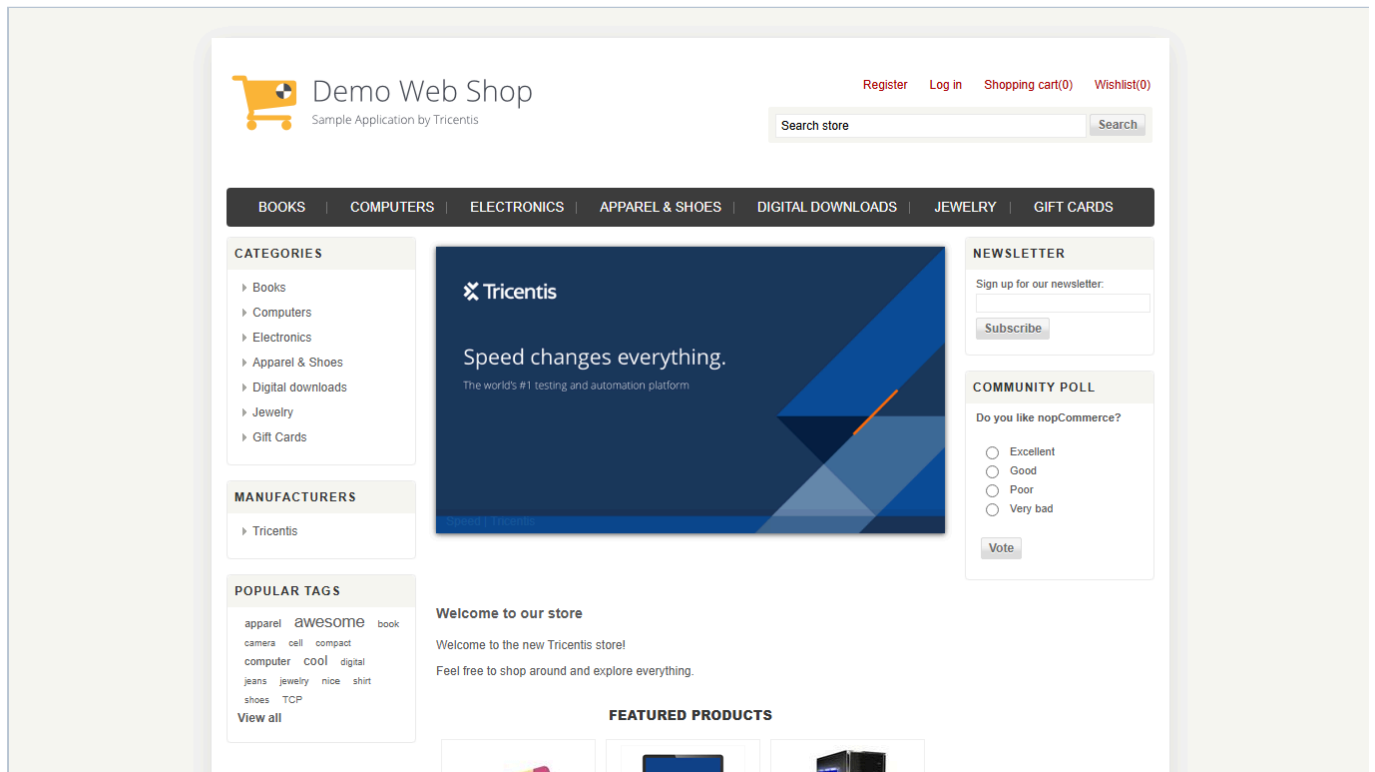
Category	Final Status
Project rebuild at new location	Completed
6 core modules	Completed
2 add-on modules	Completed and renamed clearly
Total test cases	120
Cross-browser discovery	360 tests
Execution optimization	Improved using parallel workers and reusable fixtures
Final report	Updated without code snippets
Testing plan	Available in docs folder
GitHub repository link	https://github.com/Akshit1984/Playwright_Capstone-Project

The latest test discovery command confirmed that Playwright lists 360 tests in 120 files after the add-on folder and test title renaming work.

Application Screenshots

Screenshots below provide visual evidence of important pages covered by the automation suite. These are included as supporting evidence, not as a replacement for automated test results.

Application Home Page



Application availability and featured products evidence.

Login Page



Demo Web Shop

Sample Application by Tricentis

Register

Log in

Shopping cart(0)

Wishlist(0)

Search store

Search

BOOKS

COMPUTERS

ELECTRONICS

APPAREL & SHOES

DIGITAL DOWNLOADS

JEWELRY

GIFT CARDS

CATEGORIES

» Books

» Computers

» Electronics

» Apparel & Shoes

» Digital downloads

» Jewelry

» Gift Cards

MANUFACTURERS

» Tricentis

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Welcome, Please Sign In!

New Customer

By creating an account on our website you will be able to shop faster, be up to date on an orders status, and keep track of the orders you have previously made.

Register

Returning Customer

Email:

Password:

Remember me?

Forgot password?

Log in

About login / registration

Put your login / registration information here. You can edit this in the admin site.

INFORMATION

CUSTOMER SERVICE

MY ACCOUNT

FOLLOW US

Authentication module evidence.

Product Details Page



Demo Web Shop

Sample Application by Tricentis

Register

Log in

Shopping cart(0)

Wishlist(0)

Search store

Search

BOOKS

COMPUTERS

ELECTRONICS

APPAREL & SHOES

DIGITAL DOWNLOADS

JEWELRY

GIFT CARDS

CATEGORIES

Books

Computers

Desktops

Notebooks

Accessories

Electronics

Apparel & Shoes

Digital downloads

Jewelry

Gift Cards

MANUFACTURERS

Tricentis

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Build your own computer

Build it

Availability: In stock

Free shipping

★★★★☆

378 review(s) | Add your review

Processor *

2.5 GHz Intel Pentium Dual-Core E2200 [+15.00] ▼

RAM *

2 GB ▼

HDD *

☐ 320 GB

☐ 400 GB [+100.00]

OS *

☒ Ubuntu

☐ Windows 7 [+50.00]

☐ Windows 10 [+60.00]

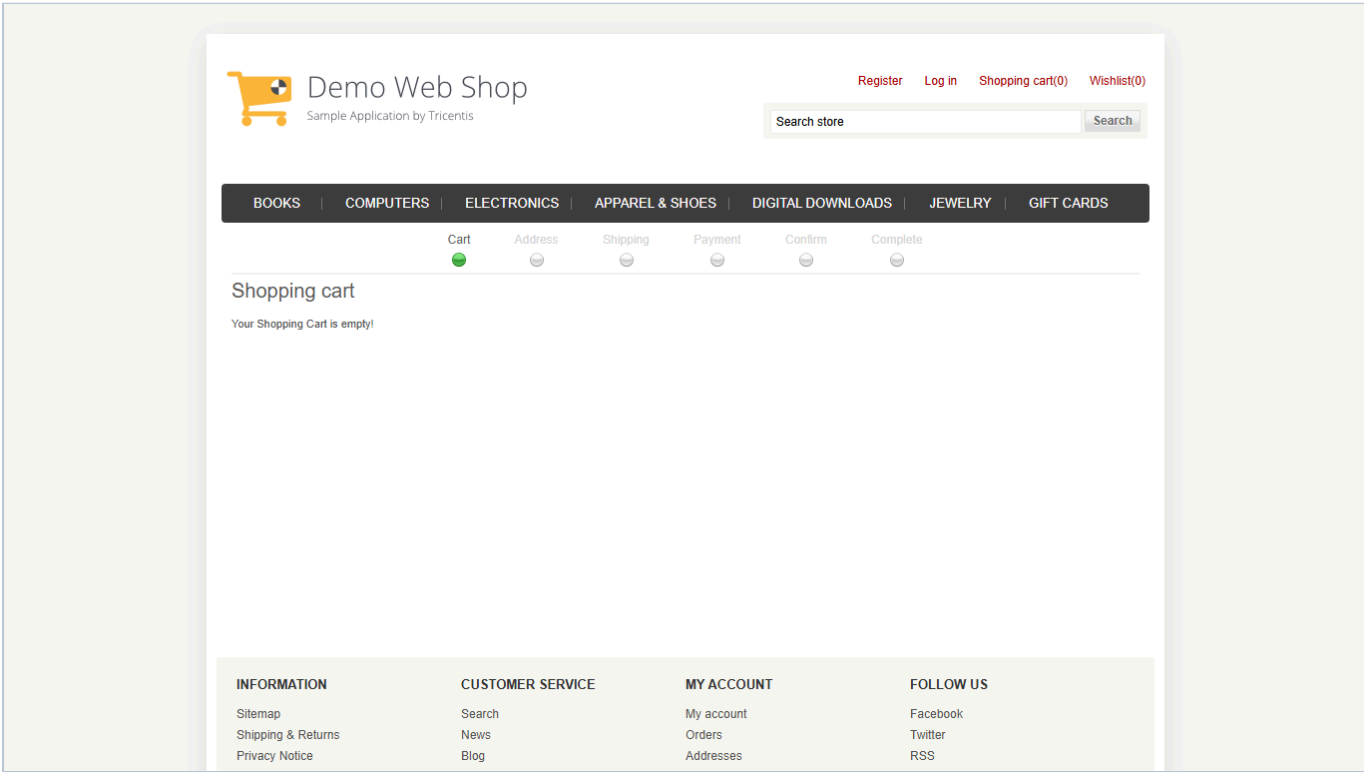
Software

☒ Microsoft Office [+50.00]

☐ Acrobat Reader [+10.00]

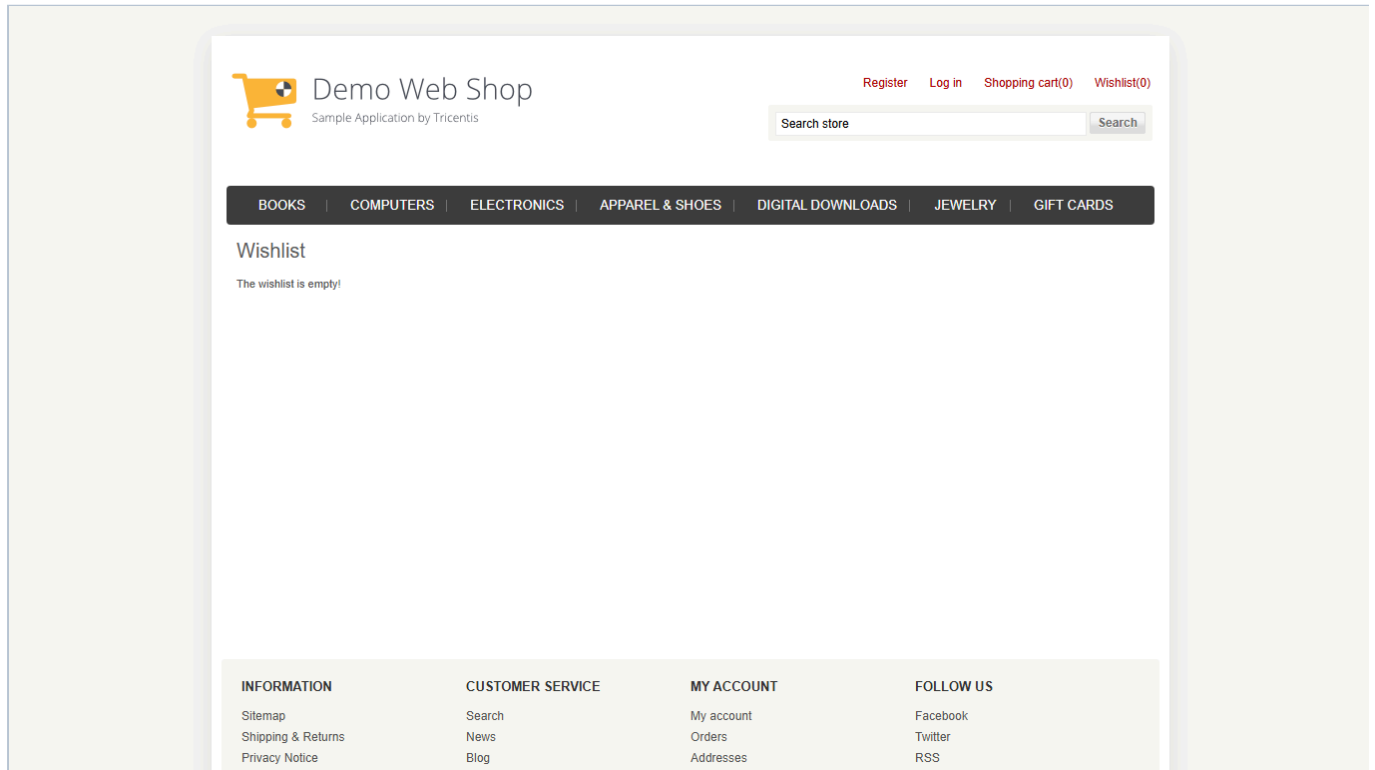
Product detail module evidence.

Shopping Cart Page



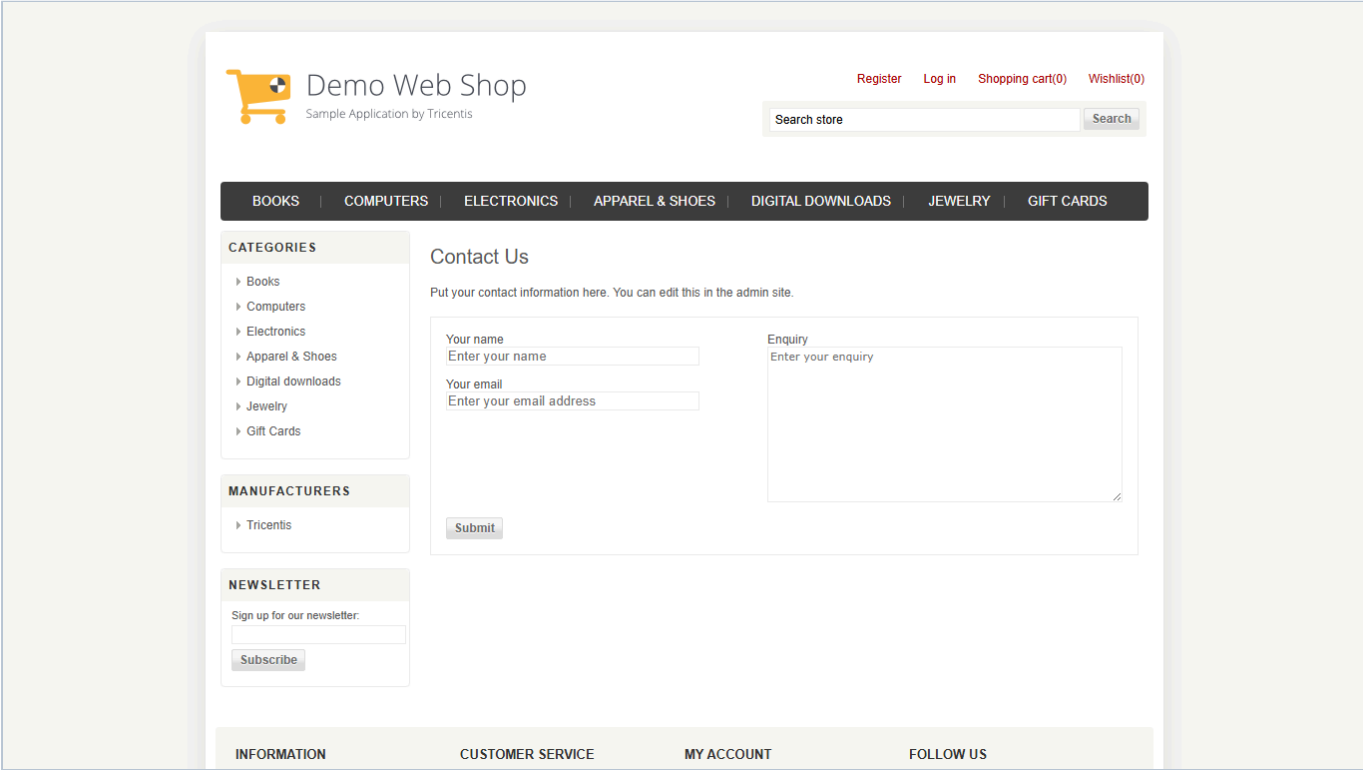
Cart and checkout module evidence.

Wishlist Add-on Page



Wishlist add-on module evidence.

Contact Us Add-on Page



Customer support add-on module evidence.

Conclusion

The capstone project successfully implements automated testing for Demo Web Shop using Playwright. The test suite is modular, traceable, and configured for cross-browser validation. It contains six required core modules and two add-on modules with clear naming across folders, files, and test output.

The documentation has been revised to follow a report-style academic structure similar to the shared sample, including overview, goals, specifications, requirements, RTM, strategy, module coverage, reporting, risks, and final summary. Code snippets have been removed as requested by the trainer.

Repository for Review: https://github.com/Akshit1984/Playwright_Capstone-Project